

STABLEMAX-PPO: FINITE-BUDGET RLHF WITH A MAX-STABILITY TRADE-OFF

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ABSTRACT

Reinforcement learning from human feedback (RLHF) has substantially improved language-model post-training, but many RLHF pipelines optimize the expected reward of a single response even when deployment selects the best from a finite set. Recent Best-of- N -aware methods reduce this mismatch, yet many operate on scalar rewards or nominal reward-model distributions, discarding the ordinal threshold structure that determines the maximum and leaving upper-tail optimization sensitive to calibration error, degradation of typical responses, and policy drift. To address these limitations, we introduce **StableMax-PPO**, which directly trains the policy for the same finite-budget maximum used at deployment while retaining the predictive distribution of a frozen ordinal reward model trained from repeated human ratings. A one-sided envelope on its CDFs yields the exact worst-case Best-of- N objective over the resulting ambiguity set, and the same threshold construction provides an exact response-level credit for PPO optimization. Nominal-mean and independent-quality floors guard against degradation of typical responses and target-evaluator over-optimization, while reference-policy KL control and state-complete rollback constrain excessive policy updates. Across three seeds, **StableMax-PPO** retains competitive finite-budget maximum performance while reducing held-out drift relative to baselines, characterizing a calibration-robust, lower-drift operating point on the max-stability frontier.

1 INTRODUCTION

Many language-model systems generate several responses and return, rerank, or inspect the best candidate, while common RLHF objectives optimize the expected score of one response. The distinction matters because a mean improvement need not increase the chance of obtaining a strong candidate within a fixed budget. Mean-PPO and GRPO remain effective general-purpose post-training methods (Shao et al., 2024; Guo et al., 2025), and recent max@ k , Best-of- N , and inference-aware policy-gradient methods better align training with finite-budget deployment (Chow et al., 2025; Tang et al., 2025; Walder & Karkhanis, 2025; Bagirov et al., 2025; Li et al., 2026; Poiani et al., 2026). We do not claim the first finite-budget objective. The unresolved setting is an imperfect ordinal evaluator: collapsing its output to a scalar discards the threshold probabilities that determine the maximum, while optimizing a nominal upper tail can amplify calibration error, sacrifice typical quality, or move the policy far from its reference. Our question is therefore how to optimize the same finite-budget statistic used at deployment while preserving the evaluator’s ordinal information and keeping quality and drift independently measurable.

We introduce StableMax-PPO, an evaluator-to-optimizer construction for this setting. A frozen ordinal Moment RM, trained from repeated HelpSteer2 helpfulness ratings (Wang et al., 2024b), represents conditional rating disagreement as a predictive distribution over $\{0, 1, 2, 3, 4\}$. The distribution retains every threshold event used by the maximum; its mean provides an ordinary model-implied quality measure, while its variance is kept as a disagreement diagnostic rather than a reward bonus. We place a one-sided calibration envelope on the CDFs in the direction adverse to upper-tail performance. The resulting product distribution exactly attains the worst-case Best-of- N value over the declared ambiguity set, and the same threshold products yield an exact response-level marginal credit for policy optimization.

This shared construction is the main technical distinction: calibration robustness changes both the deployment-level group objective and the credit that differentiates it, rather than entering as an auxiliary penalty. PPO is only the optimization carrier. A nominal-mean floor and a separately trained, frozen Quality RM from preference pairs guard typical responses and transfer beyond the target evaluator; reference-policy KL regularization and state-complete rollback control the accepted update trajectory. The robust maximum, both quality measurements, and policy drift consequently remain separate quantities, making the max–stability trade-off observable rather than hiding it in a weighted reward.

The central contribution is therefore an evaluator-to-gradient contract, rather than a new PPO optimizer in isolation or an additive list of stabilization heuristics. Existing finite-budget objectives can be optimized with a scalar reward, but that scalarization happens before the group interaction is differentiated. Here the adverse CDF envelope first defines the deployment functional and then induces the response-level credit used by PPO. Calibration robustness consequently changes both what counts as a successful group and which response receives credit, while the quality and KL mechanisms remain separately auditable.

Our contributions are:

- **A calibration-robust ordinal objective.** We derive the exact finite- N worst-case maximum over a one-sided product ambiguity set and quantify its dependence on envelope coverage and radius.
- **An exact group-to-response credit.** The same threshold construction gives a Rao–Blackwellized marginal credit whose score-function expectation is the gradient of the robust group objective and whose credits are computable in $O(N|\mathcal{R}|)$ time.
- **A controlled optimization realization.** We combine this credit with PPO while keeping nominal mean, independently evaluated quality, KL, and rollback as distinct safeguards rather than redefining the robust reward.
- **An empirical characterization of the operating frontier.** At $N = 32$ across three seeds, StableMax-PPO attains a competitive robust maximum with lower held-out drift than Mean-PPO, GRPO, and Scalar max-PPO; max-oriented comparators attain higher maxima at greater drift.

2 PROBLEM SETUP

2.1 SETTING AND ASSUMPTIONS

We study response-level RLHF with a fixed Best-of- N generation budget. The policy parameter θ is trainable, whereas the ordinal Moment RM ω , scalar Quality RM η , and reference policy π_{ref} are frozen. For $x \sim \mathcal{D}$, responses factor as $\pi_{\theta}(Y | x) = \prod_h \pi_{\theta}(a_h | x, a_{<h})$, and a rollout contains $Y_{1:N} \stackrel{\text{iid}}{\sim} \pi_{\theta}(\cdot | x)$. We assume (A1) conditional independence of rating outcomes across candidates, (A2) frozen evaluators, and (A3) the differentiability and likelihood-ratio regularity needed below. Thus changes in the measured frontier are attributable to the policy and controller rather than to a moving evaluator. Each response receives one terminal evaluation; the group statistic is formed before any response-level credit is assigned.

Assumption (A1) concerns annotation outcomes conditional on completed responses; it does not assert token independence. It is what converts each threshold event for the maximum into a product of candidate CDFs. Assumption (A2) keeps both evaluators and the reference policy fixed, so the measured frontier moves only through the policy and controller. Assumption (A3) enables the likelihood-ratio differentiation in Theorem 2. Correlated evaluator errors are therefore outside the product model rather than an effect absorbed by the optimizer.

2.2 ORDINAL EVALUATOR AND FINITE-BUDGET VALUE

The Moment RM returns $p_{\omega,r}(x, y) = \Pr_{\omega}(R = r | x, y)$ for $\mathcal{R} = \{0, 1, 2, 3, 4\}$; write $\Delta(\mathcal{R})$ for the probability simplex on these five levels. Its mean $\mu_{\omega} = \sum_r r p_{\omega,r}$ measures target-RM quality, while $\sigma_{\omega}^2 = \sum_r (r - \mu_{\omega})^2 p_{\omega,r}$ is retained only as a model-based disagreement diagnostic;

the independent scalar $Q_\eta(x, y)$ is a separate quality check. The category endpoints are fixed by the training data; the ordinal-probit parameterization is given in Appendix A. Thus the five levels are part of the evaluator interface rather than thresholds selected after observing policy outcomes. The mean, categorical vector, and variance have distinct roles: only the vector determines the finite-budget target, and the variance remains diagnostic rather than becoming a reward bonus.

For candidate j , let $F_{\omega,j}(t) = \sum_{r=0}^t p_{\omega,r}(x, Y_j)$. Under (A1), the nominal finite-support identity is $h_N^{\text{nom}} = \sum_{t=1}^4 [1 - \prod_{j=1}^N F_{\omega,j}(t-1)]$, and $J_N^{\text{nom}}(\theta) = \mathbb{E}[h_N^{\text{nom}}]$. Its threshold products show why the full ordinal vector matters: a candidate is useful when the other responses leave a threshold uncovered, which a scalar maximum cannot represent before credit assignment. The identity integrates out rating noise exactly; the top-rating event is only one of the four threshold terms. Two responses with the same predictive mean can therefore have different group value because their mass is distributed differently across thresholds.

We estimate ε_{cal} on independent prompt clusters before training (Appendix A) and use the one-sided envelope $\bar{F}_j(t) = \min\{1, F_{\omega,j}(t) + \varepsilon_{\text{cal}}\}$, with $\bar{p}_{j,r} = \bar{F}_j(r) - \bar{F}_j(r-1)$. Define $\mathfrak{U}_{\varepsilon_{\text{cal}}}(p_{\omega,1:N}) := \prod_{j=1}^N \{q_j \in \Delta(\mathcal{R}) : F_{q_j}(t) \leq \bar{F}_j(t), t = 0, \dots, 3\}$. Set $\bar{F}_j(-1) = 0, \bar{F}_j(4) = 1$. Increasing a CDF is adverse to an upper-tail maximum, so the envelope changes the deployment target and its policy credit together; it is not an auxiliary reward bonus. The radius is estimated on a split disjoint from policy training, so it fixes a declared conservatism level before policy outcomes are observed rather than tuning a reward after the fact.

2.3 ROBUST CONSTRAINED OBJECTIVE

The robust group value is $h_N^{\text{rob}} := \inf_{q_{1:N} \in \mathfrak{U}_{\varepsilon_{\text{cal}}}(p_{\omega,1:N})} \mathbb{E}_q[\max_j R_j] = \sum_{t=1}^4 [1 - \prod_{j=1}^N \bar{F}_j(t-1)]$. Our single optimization objective is

$$\begin{aligned} \max_{\theta} \quad & J_N^{\text{rob}}(\theta) := \mathbb{E}[h_N^{\text{rob}}(x, Y_{1:N})], \\ \text{s.t.} \quad & M(\theta) := \mathbb{E}[\mu_\omega(x, Y)] \geq m_0, \quad Q(\theta) := \mathbb{E}[Q_\eta(x, Y)] \geq q_0, \\ & K(\theta) := \mathbb{E}_x[\text{KL}(\pi_\theta \| \pi_{\text{ref}})] \leq \delta. \end{aligned} \quad (1)$$

The floors m_0, q_0 are fixed from the reference policy and keep target-RM and independent quality visible, while K measures policy motion separately. In PPO, the floors enter through dual credits, the KL term through the surrogate, and the distinct threshold δ_{hard} through batch rollback; the measured token KL is an acceptance statistic, not a population trust-region guarantee. This separation makes a change in the reported maximum attributable to the robust target distinguishable from a change caused by a quality multiplier or by the update-acceptance rule. In particular, the independent Quality RM is a transfer check trained from preference pairs, not another term that can be traded against the robust maximum; the multipliers only price violations of the declared floors.

3 METHOD

The method separates the deployment target from its stabilizers. The target is the robust ordinal maximum; the stabilizers control typical quality and policy motion. This separation makes the max-stability trade-off visible instead of hiding it in a scalar reward weight. The construction has two steps: first, the evaluator distribution is transformed into a worst-case finite-budget functional; second, that functional is differentiated into response-level credits and carried by PPO. The theorems below establish these steps before the implementation details introduce clipping, dual prices, and rollback.

3.1 EXACT ROBUST MAXIMUM AND MARGINAL CREDIT

Theorem 1 (Exact finite-budget robust maximum). *Under (A1), for every fixed $(x, Y_{1:N})$, an optimizer over the product ambiguity set is $q_j = \bar{p}_j$ for every j , and the attained worst-case Best-of- N value is*

$$\inf_{q_{1:N} \in \mathfrak{U}_{\varepsilon_{\text{cal}}}(p_{\omega,1:N})} \mathbb{E}_q[\max_j R_j] = \sum_{t=1}^4 \left[1 - \prod_{j=1}^N \bar{F}_j(t-1) \right]. \quad (2)$$

The key point is simultaneous optimality. The maximum is coordinatewise increasing, so saturating every admissible CDF upper bound makes all candidates stochastically smallest at once; the adversary need not choose a different perturbation for each threshold. The full ambiguity-set problem therefore reduces to four threshold products, without rating sampling or enumerating 5^N joint outcomes. This is stronger than applying a separate correction to a scalar maximum: the same envelope is feasible for the entire product set and is extremal for all thresholds simultaneously.

For a candidate i , let $\bar{R}_i \sim \bar{p}_i$ and $\bar{M}_{-i} = \max_{j \neq i} \bar{R}_j$. Its leave-one-out marginal credit is

$$q_i^{\text{rob}} := \mathbb{E}[(\bar{R}_i - \bar{M}_{-i})_+ \mid x, Y_{1:N}] = \sum_{t=1}^4 [1 - \bar{F}_i(t-1)] \prod_{j \neq i} \bar{F}_j(t-1). \quad (3)$$

The first factor is candidate i 's probability of crossing threshold t ; the product over $j \neq i$ is the probability that the other responses leave that threshold uncovered. This leave-one-out value captures incremental group credit rather than a winner indicator, so a response can matter without having the largest mean score. Prefix and suffix products compute all credits in $O(N|\mathcal{R}|)$ time per prompt using the same kernel as the group objective, without sampling latent ratings. In particular, a high-mean response can receive little marginal credit when other candidates already cover the same thresholds, while a complementary response can receive credit without being the mean-score winner. The credit therefore preserves the group interaction that would be lost by assigning each response an isolated scalar reward. It is a marginal, rather than additive, decomposition: ties or redundant candidates need not exhaust the group value, because the signal is intended to measure incremental threshold coverage.

This distinction also determines what is held fixed in the comparison. The candidate responses are sampled from the policy, but the rating randomness is integrated analytically; changing N would change the group functional, not merely the number of reward observations. At the fixed budget used here, the credit is therefore aligned with the deployment statistic at the level of both its value and its derivative.

Theorem 2 (Exact marginal policy credit). *Under (A1)–(A3), the exact group objective has the score-function gradient*

$$\nabla_{\theta} J_N^{\text{rob}}(\theta) = \mathbb{E} \left[\sum_{i=1}^N q_i^{\text{rob}} \nabla_{\theta} \log \pi_{\theta}(Y_i \mid x) \right]. \quad (4)$$

Thus the response-level estimator is unbiased before PPO clipping.

This is the evaluator-to-policy bridge: latent rating draws are integrated out, while the likelihood ratio distributes each candidate's threshold credit over its tokens. The same cached categorical probabilities define the robust target and every leave-one-out credit; PPO supplies only the clipped optimization carrier. Before clipping, no sampled rating estimator or learned group-winner critic is needed. After clipping, the estimator is an implementation of the same credit rather than a new objective; clipping and rollback are explicit controls on the accepted optimization trajectory. Thus the exact identity specifies the target gradient, while the PPO ratio, KL coefficient, and acceptance test specify only how aggressively that gradient is realized. This also clarifies the empirical attribution: a difference in the robust endpoint is a change in the evaluator-defined target, whereas a difference in held-out KL or rollback frequency reflects the controller's operating point.

3.2 CONSTRAINED PPO UPDATE

For dual variables $\lambda_m, \lambda_q \geq 0$, the response-level credit used by PPO is

$$\tilde{q}_i = q_i^{\text{rob}} + \frac{\lambda_m}{N} \mu_{\omega}(x, Y_i) + \frac{\lambda_q}{N} Q_{\eta}(x, Y_i), \quad A_i = N[\tilde{q}_i - b_{\psi}(x)]. \quad (5)$$

The factors $1/N$ match the one-response constraints to the group objective; the prompt-only baseline preserves the score-function identity. We use the standard clipped PPO surrogate and full-categorical token KL. Writing $\rho_{i,h} = \pi_{\theta}(a_{i,h} \mid s_{i,h}) / \pi_{\text{old}}(a_{i,h} \mid s_{i,h})$, the update maximizes the standard per-token expression $\min\{\rho_{i,h} A_i, \text{clip}(\rho_{i,h}, 1 - \epsilon_{\text{PPO}}, 1 + \epsilon_{\text{PPO}}) A_i\}$ averaged over candidates, minus $\beta_{\text{KL}} \widehat{\text{KL}}$, and divides by the lagged RMS scale. The full implementation, including projected dual updates, is specified in Appendix C.

Before a tentative step, StableMax-PPO snapshots all policy parameters and the complete optimizer state. If the post-update KL on the same rollout states exceeds δ_{hard} , both snapshots are restored; baseline, dual, RMS, and controller states advance only after acceptance. Snapshotting the optimizer moments is necessary for Adam-like optimizers, and the resulting rollback count is comparable across baselines under the shared controller. The soft target δ and the hard threshold δ_{hard} serve different purposes: the former is the reference-policy constraint represented by the KL controller, whereas the latter is a batch-level reject rule. An accepted update can still differ from the population-constrained optimum, which is why held-out KL is measured independently rather than inferred from the rollback threshold.

3.3 STABLEMAX-PPO

Here Opt denotes the complete policy-optimizer state, including momentum and adaptive-variance buffers. We compute $\widehat{\text{KL}}$ as the batch mean of the full-categorical token KL on rollout states, not from sampled actions. The update uses q_i^{rob} and A_i from eqs. (3) and (5). Starting from the current policy and controller state, each iteration evaluates only the frozen models on a fixed-size rollout; the policy and controller are updated, whereas the models, calibration radius, floors, soft target δ , and hard rollback threshold are fixed by the protocol.

Algorithm 1: StableMax-PPO update

Input: Prompts $x \sim \mathcal{D}$; initial trainable policy π_θ ; frozen $p_\omega, Q_\eta, \pi_{\text{ref}}$; fixed $N, \varepsilon_{\text{cal}}, m_0, q_0, \delta, \delta_{\text{hard}}$.

Output: Accepted policy parameters and optimizer/controller state.

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1 Initialize Opt,  $b_\psi, \lambda_m, \lambda_q, \beta_{\text{KL}}$ , and lagged RMS scale  $s$ ;
2 for  $k = 1, \dots, T_{\text{pol}}$  do
3   Set  $\pi_{\text{old}} \leftarrow \pi_\theta$  and roll out  $N$  responses per prompt; evaluate the frozen  $p_\omega, Q_\eta$ ;
4   Compute  $q_i^{\text{rob}}$  from eq. (3) and  $A_i$  from eq. (5);
5   Snapshot  $(\theta, \text{Opt})$  and take one clipped-PPO step with coefficient  $\beta_{\text{KL}}$ ;
6   Measure  $\widehat{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})$  on the same rollout states;
7   if  $\widehat{\text{KL}} > \delta_{\text{hard}}$  then
8     Restore  $(\theta, \text{Opt})$ , discard the transition, and do not advance  $b_\psi, \lambda_m, \lambda_q, \beta_{\text{KL}}, s$ ;
9   else
10    Commit the transition; fit  $b_\psi$ , project the dual updates, and update  $\beta_{\text{KL}}, s$  from accepted data;
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All quantities needed to reproduce the update are fixed above or in Appendix C; only the policy and controller states are updated, and rejected transitions advance none of them.

4 THEORETICAL ANALYSIS

The previous section gives the exact target and its policy-gradient bridge. Here we state the guarantees that explain the role of the calibration radius and the update controller. Proofs are in Appendix A.

Theorem 3 (Calibration-robust lower bound). *Under (A1), let F_j^* be the true conditional CDF. If $F_j^*(t) \leq \bar{F}_j(t)$ for every candidate and $t \in \{0, 1, 2, 3\}$, then the true conditional maximum satisfies $\mathbb{E}_*[\max_j R_j \mid x, Y_{1:N}] \geq h_N^{\text{rob}}(x, Y_{1:N})$. Without exact coverage, $\mathbb{E}_*[\max_j R_j \mid x, Y_{1:N}] \geq h_N^{\text{rob}}(x, Y_{1:N}) - 4N\xi(x, Y_{1:N})$, where $\xi = \max_{j \in [N], t \in \{0, 1, 2, 3\}} [F_j^*(t) - \bar{F}_j(t)]_+$.*

This is a conditional envelope statement, not a distribution-free guarantee: cluster calibration chooses the radius but does not establish pointwise coverage for every generated response. If the envelope misses by at most ξ , the four ordinal thresholds and N candidate CDFs contribute the explicit $4N\xi$ degradation term. The factor depends on the declared five-level support and product model, so it is a sensitivity certificate for this evaluator interface rather than a claim of universal calibration.

The factor $4N$ follows from the representation itself: at each of the four nontrivial thresholds, changing one candidate CDF changes the corresponding product by at most its CDF error. Summing these

N -candidate sensitivities over thresholds gives the stated degradation, independently of the optimizer.

The radius also has a deterministic target-level interpretation independent of how it is estimated.

Proposition 1 (Calibration-radius monotonicity). *For $0 \leq \varepsilon_1 \leq \varepsilon_2 \leq 1$, $0 \leq J_N^{\text{rob}}(\theta; \varepsilon_1) - J_N^{\text{rob}}(\theta; \varepsilon_2) \leq 4N(\varepsilon_2 - \varepsilon_1)$, and at $\varepsilon = 0$, $J_N^{\text{rob}}(\theta; 0) = J_N^{\text{nom}}(\theta)$.*

The radius is therefore an interpretable target-level knob: increasing it makes the robust value more conservative with bounded sensitivity. Unlike a KL coefficient, it changes both the group target and its marginal credit, so the nominal-EV comparator isolates a change in the evaluator-defined objective. A radius change is consequently not equivalent to retuning PPO: it changes the policy-gradient direction as well as the objective level.

This gives the calibration parameter a role that can be inspected separately from policy regularization. Setting $\varepsilon_{\text{cal}} = 0$ recovers the nominal ordinal objective exactly, while increasing it cannot improve the robust value by construction. The sensitivity bound also prevents a small reported change from being interpreted as a new optimization effect when it is only a change in the declared uncertainty set.

Let $\phi_i = \nabla_{\theta} \log \pi_{\theta}(Y_i | x)$. Consider a hypothetical estimator that samples latent ratings from the robust envelope, $G_{\text{samp}} = \sum_i (\bar{R}_i - \bar{M}_{-i}) \phi_i$, and the implemented credit estimator $G_{\text{RB}} = \sum_i q_i^{\text{rob}} \phi_i$. The former introduces evaluator-sampling noise that is unnecessary once the categorical probabilities are available.

Proposition 2 (Rao–Blackwell variance dominance). *Conditioned on $(x, Y_{1:N})$, $G_{\text{RB}} = \mathbb{E}[G_{\text{samp}} | x, Y_{1:N}]$. Consequently, for every vector v , $\text{Var}(v^{\top} G_{\text{RB}}) \leq \text{Var}(v^{\top} G_{\text{samp}})$.*

The exact credit removes variance from drawing latent ordinal ratings while leaving rollout and policy-sampling variance unchanged. The PPO update thus uses the conditional expectation of the group credit without extra evaluator calls; the proposition makes no claim that total optimization variance vanishes. It identifies one avoidable source of noise and leaves response-sampling, optimization, and controller variability to the empirical analysis.

The distinction matters because the evaluator already supplies the full categorical probabilities. Sampling an ordinal label would add Monte Carlo noise without adding information about the declared objective; the exact credit removes that noise while preserving the same group interaction and score-function expectation.

The last result concerns the realized optimizer state rather than the population objective.

Proposition 3 (Rollback correctness). *If a tentative update is rejected, restoring the parameter and optimizer snapshots exactly returns the measured rollout-state policy and optimizer to their pre-update values. This is a batch-level invariant, not a population KL guarantee.*

The envelope protects the model-conditional target, the marginal identity makes it optimizable, and rollback controls the measured batch update. Restoring the complete snapshot is essential for optimizer and controller consistency, but does not imply a population trust region after an accepted update; maximum, quality, held-out KL, and rollback frequency are therefore reported separately. In particular, restoring policy weights without adaptive buffers could make a rejected transition affect the next step, so the invariant is stated for the complete optimizer/controller state.

The invariant is local to the measured rollout and to the state that the controller can mutate. It guarantees that a rejected step contributes neither policy weights nor hidden optimizer momentum to the next accepted step, but it does not constrain later accepted updates or imply a global trust region. This is why the empirical report includes both rejection frequency and held-out KL.

5 EXPERIMENTS

Our empirical question is whether the evaluator-to-optimizer contract produces a reproducible maximum stability operating point. We hold the response budget, prompt stream, seeds, optimizer, and rollback rule fixed, while comparators vary the deployment functional, calibration treatment, or safeguard. We report robust maximum together with independent quality, held-out KL, and rollback frequency so that target performance and policy stability remain separately observable.

5.1 EXPERIMENTAL SETUP

The policy is Qwen2.5-1.5B-Instruct. The frozen ordinal Moment RM is a Qwen2.5-14B model trained on HelpSteer2 repeated helpfulness ratings, with the decontaminated preference replay specified in Appendix B (Wang et al., 2024b). The independent scalar Quality RM is a separately trained Qwen2.5-14B model fit on disjoint preference data. We use three confirmatory seeds {314, 2718, 1618}, $N = 32$, and 512 held-out prompts per seed. Primary baselines are Mean-PPO + Q, GRPO + Q, and Scalar Max@32-PPO + Q; full settings, ablations, and audit records are in Appendix B. We report robust maximum, held-out KL, and rollback frequency because they measure the two sides of the proposed operating point.

Repeated helpfulness ratings make the categorical output and its disagreement diagnostic identifiable; the independent Quality RM and generated-output proxies test transfer beyond the target evaluator. Shared prompts, seeds, generation budgets, and rollback thresholds keep every comparison on the same response-sampling problem. All confirmatory claims therefore concern this fixed $N = 32$ operating point, not scale or budget generalization.

The primary comparators are chosen to isolate the two axes in the objective. Mean-PPO keeps the conventional single-response target, GRPO keeps a group-relative scalar target, and Scalar max-PPO uses the same finite budget with the ordinal distribution collapsed to response means. Matching their optimizer, quality safeguards, and rollback rule makes the comparison about the deployment functional and its credit assignment, rather than about a different amount of computation or a different update stabilizer. The additional evaluator and floor variants in Appendix B then vary one contract component at a time.

5.2 MAIN RESULTS

StableMax-PPO reaches a robust expected maximum of 3.458 and held-out KL of 0.034, compared with 3.424 and 0.070 for Mean-PPO. Under the shared rollback controller it rejects 31.0% of tentative updates, versus 58.6% for Mean-PPO and 82.3% for GRPO. GRPO and Scalar max-PPO attain higher robust maxima, so the evidence supports a lower-drift operating point rather than universal dominance. Because the primary endpoint is the same h_{32}^{rob} functional used in training, this comparison directly tests finite-budget objective alignment rather than evaluating a mean-trained policy only after Best-of-32 selection.

Figure 1 makes the trade-off explicit: StableMax-PPO has lower held-out KL than all three trainable primary comparators while retaining a competitive robust maximum, whereas the max-oriented comparators remain higher on that endpoint. Rollback frequency is an optimization diagnostic, not a population guarantee.

The aggregate numbers quantify the operating point rather than a universal ranking. Relative to Mean-PPO, StableMax-PPO increases robust maximum by 0.034 while reducing held-out KL from 0.070 to 0.034 (a 51% relative reduction). Relative to Scalar max-PPO, it gives up 0.059 robust-max points while using 0.020 less KL. These are the two axes that the method exposes; collapsing them into one weighted score would conceal the engineering trade-off the experiment is designed to measure.

The comparison should be read as an operating-point result, not as evidence that the safeguards make every endpoint larger. GRPO and Scalar max-PPO demonstrate that more aggressive max-seeking can reach a higher target-RM maximum under this protocol, but their larger held-out KL records the additional policy movement. StableMax-PPO instead reaches a nearby point with lower drift, which

Table 1: Primary aggregate results (three seeds, 512 prompts per seed).

Method	Robust max	95% CI	Held-out KL	Rollbacks/900
StableMax-PPO	3.458	[3.411, 3.506]	0.034	279 (31.0%)
Mean-PPO	3.424	[3.354, 3.517]	0.070	527 (58.6%)
GRPO	3.555	[3.530, 3.578]	0.094	741 (82.3%)
Scalar max-PPO	3.517	[3.476, 3.558]	0.054	285 (31.7%)
Best-of-32 (base)	3.086	[3.045, 3.125]	–	–

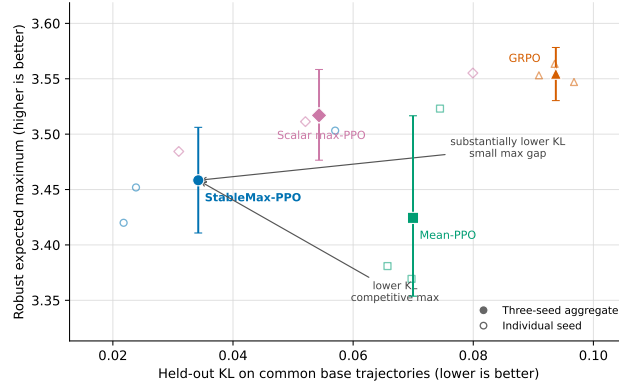


Figure 1: Robust maximum versus held-out KL. Filled markers are three-seed aggregates; open markers are individual seeds and bars are 95% intervals.

is the behavior predicted by treating quality and motion as constraints rather than as unreported side effects.

5.3 ABLATION AND STABILITY ANALYSIS

The component variants isolate changes to the target from changes to the controller. Removing the mean floor raises the aggregate robust maximum from 3.458 to 3.508, but also raises held-out KL from 0.034 to 0.048. Removing the independent-quality floor gives the operating point (3.467, 0.048), while removing calibration robustification gives (3.421, 0.028), in (robust maximum, held-out KL) coordinates. The three-seed intervals overlap and the per-seed effects are heterogeneous, so these comparisons are sensitivity analyses rather than isolated causal effects. In particular, removing calibration changes the evaluator-defined target, whereas removing a floor changes the accepted-update trajectory. The complete aggregate and per-seed tables are in Appendix B; no component is presented as a universally improving reward term.

Together, the results answer the objective-design question at the evaluated operating point: the robust ordinal credit trains against the same finite-budget statistic used for evaluation, while quality floors and held-out KL keep typical performance and policy movement visible. Independent generated-output diagnostics are reported in Appendix B.9; they are model-based diagnostics rather than fresh human judgments.

6 LIMITATIONS

Scope. The endpoint is induced by frozen evaluators, with no fresh human labels for generated outputs; model proxies may disagree with people, the 1.5B/14B capacity gap may induce evaluator shift, and three seeds do not establish scale or domain generalization. We fix $N = 32$, and the exact product form assumes conditionally independent rating outcomes, so correlated annotation or evaluator errors are outside the guarantee.

Future work. Blinded human ratings, additional response budgets and domains, larger policies, and correlated-error ambiguity sets are natural next tests. The last direction is especially important because it asks whether the threshold construction can retain tractable policy credit when evaluator errors are shared across candidates.

Reproducibility statement. The supplementary artifact contains the source, official ICLR style files, protocol, result tables, verification JSON, build commands, and the anonymous repository at <https://github.com/LoveGuys-cmd/rlhf-artifact>. The main paper specifies the objective, algorithm, seeds, metrics, and aggregate results; appendices give proofs and the complete protocol. Model weights, credentials, private data, and other non-redistributable files are excluded.

AI use statement. Generative AI tools assisted with paper organization, methodological exposition, language editing, LaTeX and figure formatting, and debugging of analysis code. The authors in-

independently checked all mathematical claims and proofs, verified implementation and numerical values against frozen result files, and reviewed every AI-assisted change. No AI system generated experimental data or replaced author verification.

Ethics statement. The study uses model-generated text and existing preference data. It collects no new human-subject data or personal information. Model-based preference proxies are reported as proxies, not as human judgments of newly generated outputs; possible reward-model exploitation and evaluator shift are disclosed in Section 6 and Appendix B.

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Appendix roadmap. Appendix A gives the ordinal parameterization and proofs of the exact objective, calibration, gradient, variance, and rollback claims. Appendix B records the complete experimental protocol and diagnostics; Appendix C gives implementation conventions for Algorithm 1; machine-readable hashes and verification records are summarized at the end of Appendix C.

A DETAILED THEORETICAL ANALYSIS

A.1 ORDINAL-PROBIT PARAMETERIZATION

Write $\Delta(\mathcal{R})$ for the probability simplex on $\mathcal{R} = \{0, 1, 2, 3, 4\}$. The frozen Moment RM emits a latent location $m_\omega(x, y)$ and positive scale $s_\omega(x, y)$. With frozen cutpoints $c_1 < \dots < c_4$, its five-category probabilities are $p_{\omega,0} = \Phi((c_1 - m_\omega)/s_\omega)$, $p_{\omega,r} = \Phi((c_{r+1} - m_\omega)/s_\omega) - \Phi((c_r - m_\omega)/s_\omega)$ for $r = 1, 2, 3$, and $p_{\omega,4} = 1 - \Phi((c_4 - m_\omega)/s_\omega)$. The moments used by the policy are $\mu_\omega = \sum_{r=0}^4 r p_{\omega,r}$ and $\sigma_\omega^2 = \sum_{r=0}^4 (r - \mu_\omega)^2 p_{\omega,r}$. Repeated-rating likelihood and disagreement supervise the conditional scale; therefore this variance is a model-based signal of annotation disagreement, not epistemic uncertainty or a stand-alone reward. Proper scoring and calibration are required because the complete categorical distribution enters the finite-budget objective (McCullagh, 1980; Kendall & Gal, 2017; Lou et al., 2024; Sun et al., 2025; Gneiting & Raftery, 2007; Guo et al., 2017).

This parameterization is included to make the evaluator interface reproducible, not to claim a new ordinal-link model. The methodological choice is what the policy does with the frozen output: the same five probabilities support the nominal mean, threshold products, robust envelope, and disagreement diagnostic. Keeping these uses separate is important because replacing the vector by its mean would make the finite-budget objective and its calibration check unidentifiable.

Proposition 4 (Finite-support threshold decomposition). *For conditionally independent ratings supported on $\mathcal{R} = \{0, 1, 2, 3, 4\}$, the identity $h_N^{\text{nom}} = \sum_{t=1}^4 [1 - \prod_{j=1}^N F_{\omega,j}(t-1)]$ is exact, lies in $[0, 4]$, and is nondecreasing under first-order stochastic improvement of any candidate. Its highest-threshold component is $1 - \prod_j (1 - p_{\omega,4}(x, Y_j))$.*

The proposition is the finite-support sanity check for the main construction. It also explains why the top-rating diagnostic in the artifact is not a substitute for the reported objective: it is only the $t = 4$ term, whereas the method trains against all four threshold events. This distinction prevents a policy from appearing successful by concentrating probability only at the highest category while degrading the intermediate ordinal structure.

Proof. For any integer-valued $Z \in \{0, \dots, 4\}$, the tail-sum identity gives $\mathbb{E}[Z] = \sum_{t=1}^4 \Pr\{Z \geq t\}$. Taking $Z = \max_j R_j$ and using (A1) gives $\Pr\{\max_j R_j < t \mid x, Y_{1:N}\} = \prod_{j=1}^N F_{\omega,j}(t-1)$.

Substitution proves the stated nominal identity. Each term lies in $[0, 1]$, so the sum lies in $[0, 4]$. First-order stochastic improvement decreases a candidate CDF and therefore cannot decrease any summand. At $t = 4$, $F_{\omega,j}(3) = 1 - p_{\omega,4}(x, Y_j)$, giving the highest-threshold expression above. $\square$

Proposition 5 (Prompt-cluster calibration statement). *Assume that the validation prompt clusters are independent and that the reward model is fixed before they are evaluated. For cluster g , let $R_{g\ell}$ denote its repeated ratings, n_g its number of ratings, and $F_{\omega,g\ell}(t)$ the frozen evaluator CDF at threshold t . Define $D_{g,t} = n_g^{-1} \sum_{\ell=1}^{n_g} \{1[R_{g\ell} \leq t] - F_{\omega,g\ell}(t)\}$ for $t = 0, \dots, 3$, and let $\hat{d}_t = G_{\text{cal}}^{-1} \sum_{g=1}^{G_{\text{cal}}} D_{g,t}$, where G_{cal} is the number of independent validation clusters. Let $\mathbb{E}_g$ denote expectation over the validation-cluster distribution. Define the calibration radius as*

$$\varepsilon_{\text{cal}} = \min \left\{ 1, \max_{t=0,\dots,3} (\hat{d}_t)_+ + \sqrt{\frac{\log(4/\alpha_{\text{cal}})}{2G_{\text{cal}}}} \right\}. \quad (6)$$

Then, with probability at least $1 - \alpha_{\text{cal}}$,

$$\mathbb{E}_g[D_{g,t}] \leq \varepsilon_{\text{cal}} \quad \text{simultaneously for } t = 0, 1, 2, 3.$$

Proof. Conditional on the frozen responses and predictions, changing all binary indicators in a prompt cluster from zero to one changes $D_{g,t}$ by at most one. Thus each cluster residual lies in an interval of width one. Hoeffding's inequality gives $\Pr(\mathbb{E}_g[D_{g,t}] - \hat{d}_t > \sqrt{\log(4/\alpha_{\text{cal}})/(2G_{\text{cal}})}) \leq \alpha_{\text{cal}}/4$.

A union bound over four thresholds proves the simultaneous event. On that event, every right-hand side is bounded by the unclipped expression in eq. (6). If clipping is active, $\varepsilon_{\text{cal}} = 1$, while $\mathbb{E}_g[D_{g,t}] \leq 1$, so the claim still holds. $\square$

Scope of the calibration proposition. Proposition 5 is a population-average statement over the validation prompt distribution, not a distribution-free pointwise guarantee for every generated response. The calibration split fixes the radius before training, whereas the main theorem is conditional on its envelope covering a particular candidate; the artifact records these two claims separately.

Proof of Theorem 1. The envelope $\bar{F}_j$ is nondecreasing, has boundary values zero and one, and therefore defines a valid categorical distribution. For every feasible $q_j \in \mathcal{U}$, we have

$$F_{q_j}(t) \leq \bar{F}_j(t), \quad t = 0, 1, 2, 3.$$

Hence q_j first-order stochastically dominates $\bar{p}_j$, or equivalently $\bar{p}_j$ is the stochastically smallest distribution in the ambiguity set. Since the maximum is coordinatewise nondecreasing in its ratings, replacing every feasible candidate distribution by $\bar{p}_j$ cannot increase its expectation. The envelope itself is feasible and thus attains the infimum. Applying Proposition 4 gives the stated closed form. $\square$

Proof of Theorem 2. For nonnegative integer ratings, the positive-part marginal credit can be decomposed as

$$(\bar{R}_i - \bar{M}_{-i})_+ = \sum_{t=1}^4 \mathbf{1}\{\bar{R}_i \geq t, \bar{M}_{-i} < t\}.$$

Taking conditional expectations and using the conditional independence assumption (A1) gives

$$\Pr\{\bar{R}_i \geq t, \bar{M}_{-i} < t \mid x, Y_{1:N}\} = [1 - \bar{F}_i(t-1)] \prod_{j \neq i} \bar{F}_j(t-1),$$

which proves Equation (3).

For the gradient identity, fix i and condition on (x, Y_{-i}) . The robust group value decomposes as

$$h_N^{\text{rob}} = \mathbb{E}[\bar{M}_{-i} \mid x, Y_{-i}] + q_i^{\text{rob}}.$$

The first term is independent of Y_i , and the score-function identity implies that its product with $\nabla_{\theta} \log \pi_{\theta}(Y_i | x)$ has conditional expectation zero. Applying the likelihood-ratio derivative to all N i.i.d. responses and summing the remaining terms proves Equation (4). $\square$

Proof of Theorem 3. Under (A1), let F_j^* denote the true conditional CDF. Under the pointwise coverage premise $F_j^*(t) \leq \bar{F}_j(t)$ for every candidate and threshold, first-order stochastic dominance directly yields the lower-bound inequality stated in Theorem 3. The same independence assumption permits the product comparison used for the misspecified case.

For the misspecified case, define

$$\xi \triangleq \max_{j,t} [F_j^*(t) - \bar{F}_j(t)]_+.$$

At each threshold, the true and robust tail-sum terms can differ only through the corresponding CDF products. For arbitrary $a_j, b_j \in [0, 1]$, the product difference is bounded by the sum of positive differences:

$$\prod_j a_j - \prod_j b_j \leq \sum_j (a_j - b_j)_+.$$

Applying this inequality with $a_j = F_j^*(t-1)$ and $b_j = \bar{F}_j(t-1)$ at each of the four thresholds, the discrepancy is at most $N\xi$ per threshold, and therefore at most $4N\xi$ after summation. Taking outer expectations preserves the resulting inequality:

$$\mathbb{E}_*[\max_j R_j] \geq h_N^{\text{rob}} - 4N\xi.$$

$\square$

Proof of Proposition 1. Fix a candidate group and threshold t , and define

$$a_j(\varepsilon) \triangleq \min\{1, F_{\omega,j}(t-1) + \varepsilon\}.$$

For $0 \leq \varepsilon_1 \leq \varepsilon_2 \leq 1$, the clipping map is nondecreasing and 1-Lipschitz, which gives

$$0 \leq a_j(\varepsilon_2) - a_j(\varepsilon_1) \leq \varepsilon_2 - \varepsilon_1.$$

Using the telescoping product identity and the fact that $a_j(\varepsilon) \in [0, 1]$, we have

$$0 \leq \prod_j a_j(\varepsilon_2) - \prod_j a_j(\varepsilon_1) \leq \sum_j (a_j(\varepsilon_2) - a_j(\varepsilon_1)) \leq N(\varepsilon_2 - \varepsilon_1).$$

Substituting this bound into the four-term tail-sum expression for h_N^{rob} , and then taking expectations over prompts and responses, proves Proposition 1.

At $\varepsilon = 0$, the robust envelope equals the nominal CDF, so the exact infimum in Theorem 1 equals $J_N^{\text{nom}}(\theta)$. $\square$

Proof of Proposition 2. By the conditional credit definition in Equation (3), we have

$$\mathbb{E}[(\bar{R}_i - \bar{M}_{-i})_+ | x, Y_{1:N}] = q_i^{\text{rob}},$$

for every i . The score terms ϕ_i are fixed under this conditioning, so linearity gives

$$\mathbb{E}[G_{\text{samp}} | x, Y_{1:N}] = G_{\text{RB}}.$$

For any fixed vector v , the law of total variance yields

$$\text{Var}(v^{\top} G_{\text{samp}}) = \mathbb{E}[\text{Var}(v^{\top} G_{\text{samp}} | x, Y_{1:N})] + \text{Var}(v^{\top} G_{\text{RB}}),$$

which directly implies the stated variance inequality. $\square$

Proof of Proposition 3. Let (θ_0, Opt_0) denote the snapshot taken before a tentative update. If the measured rollout-state KL exceeds δ_{hard} , the algorithm executes the restoration $(\theta, \text{Opt}) \leftarrow (\theta_0, \text{Opt}_0)$. By the semantics of the implementation's state-copy operation, both objects are restored exactly to their pre-update values. Because controller-state updates occur only in the accepted branch, a rejected transition also leaves the baseline, dual variables, KL coefficient, and RMS scale unchanged. This statement is conditional on the measured batch and therefore makes no population-level KL claim. $\square$

Proposition 6 ($N = 1$ reduction and complexity). For $N = 1$,

$$h_1^{\text{rob}} = q_1^{\text{rob}} = \sum_{r=0}^4 r \bar{p}_{1,r}.$$

Thus the upper-tail reward component reduces exactly to the robust mean reward. For general N , all group values and marginal credits can be computed in $O(N|\mathcal{R}|)$ time using threshold-wise prefix and suffix products.

This reduction serves as a useful implementation sanity check: at $N = 1$, the method must recover the robust mean and cannot manufacture a group advantage. For $N > 1$, the additional terms exactly capture the response’s incremental chance of opening a new threshold above the other candidates. The test suite validates both regimes, ensuring that any improvement in the group objective cannot be attributed to a special-case implementation artifact.

Proof. With a single candidate, the empty opponent maximum is zero by convention, so $(\bar{R}_1 - 0)_+ = \bar{R}_1$, and the first equality follows directly. For the complexity analysis, note that evaluating each of the four nontrivial thresholds requires one forward and one backward product scan over the N candidates. Hence the total time is $O(N \cdot |\mathcal{R}|)$, where $|\mathcal{R}| = 5$. $\square$

B FULL EXPERIMENTAL PROTOCOL

B.1 EXPERIMENTAL QUESTIONS

The experiments are designed to answer four predeclared questions:

1. Does StableMax-PPO improve the held-out robust $\mathbb{E}[\max_{j \leq 32} R_j]$ objective relative to mean optimization and a direct scalar max@32 policy gradient?
2. Are gains preserved under the nominal exact ordinal maximum and the top-rating probability $\Pr\{\exists j : R_j = 4\}$?
3. Do nominal expected rating, independent Quality RM score, and reference-policy KL remain within their predeclared safeguards?
4. Do independent generated-output proxies and existing human-labeled preference diagnostics provide evidence against target-RM exploitation?

No result is used to remove a comparator, redefine a threshold, or select a more favorable protocol.

B.2 PROTOCOL AUDIT AND SCIENTIFIC INTERPRETATION

The artifact verifier passed for seeds 314, 2718, and 1618. The terminal record preserves every predeclared proxy, non-inferiority, safeguard, and KL outcome for auditability. We interpret these fields as protocol diagnostics that delimit the measured max–stability frontier, not as a universal method ranking or a proof that reward-model exploitation is absent.

The diagnostic vector is read as a decision record rather than a single leaderboard score. A method can satisfy KL and quality safeguards while losing a max comparison, or improve the target reward while degrading an independent proxy. Reporting the vector preserves these distinctions and prevents a favorable endpoint from hiding a failure mode that the method was designed to measure.

The frozen ordinal calibration report uses the simultaneous one-sided level $\alpha_{\text{cal}} = 0.05$ and records $\varepsilon_{\text{cal}} = 0.0468083$. Its prompt-cluster count and hash are preserved in the machine-readable calibration report consumed by the training and verification scripts; the value is fixed before any policy run.

B.3 MODELS, DATA, AND SPLIT DISCIPLINE

The policy is Qwen2.5-1.5B-Instruct (Yang et al., 2024). The frozen Moment RM is a Qwen2.5-14B ordinal model trained on HelpSteer2 repeated helpfulness ratings, with the decontaminated preference replay recorded in its training manifest (Wang et al., 2024b). It exports $p_{\omega,0:4}$ directly; exposing

the category probabilities does not retrain or alter its accepted checkpoint. An independently trained Qwen2.5-14B scalar Quality RM, fit separately on disjoint preference data, supplies Q_η and has no variance head. Its frozen 4,096-pair confirmation accuracy is 94.60% with a Wilson lower 95% bound of 93.87%.

All policy, reward-model, evaluator, and lockbox splits are content-hashed. Before partitioning RewardBench, we remove prompt and response overlap with the policy and Quality-RM data (Lambert et al., 2024). The remaining calibration, policy-diagnostic, and reserve partitions are also mutually prompt- and response-disjoint. They contain 1,024, 1,024, and 427 pairs, respectively. The sealed Skywork diagnostic contains 4,096 pairs and is excluded from Quality-RM training and selection (Liu et al., 2024).

The Moment RM and policy prompts remain drawn from the same broad HelpSteer2 domain. Disjoint rows prevent direct leakage but do not establish out-of-domain generalization; this is treated as a limitation rather than hidden by the split terminology.

B.4 FROZEN TRAINING PROTOCOL

The primary response budget is $N = 32$. Each trainable confirmatory method uses three seeds {314, 2718, 1618}, 300 policy steps, two prompts per step, 32 responses per prompt, learning rate 2×10^{-5} , one PPO epoch, clip 0.2, LoRA rank 8 with scale 32, sampling temperature 1, $\text{top}_p = 1$, maximum prompt length 384, and maximum response length 64. Seed 42 is used only for development and real-GPU smoke tests and is excluded from all confirmatory intervals, gates, and aggregate claims. The target KL is 0.02 and the hard rollback level is 0.04. All methods share the same prompt stream, generation budget, optimizer class, policy architecture, and checkpoint-selection rule.

The non-inferiority floors are frozen before any trainable policy outcome is observed. The reference sample contains 2,048 base-policy responses from 512 independent policy-training prompts, with four responses per prompt. For $Z \in \{\mu_\omega, Q_\eta\}$, let $\bar{Z}_g$ be the four-response prompt mean and let L_Z be the fifth percentile of 10,000 bootstrap means obtained by resampling prompts. The frozen floor is $z_0 = L_Z - 0.1s_Z$, where s_Z is the response-level standard deviation in the same reference sample. The bootstrap unit, one-sided level, 10,000 draws, sample counts, generation seed, bootstrap seed, and practical margin are all predeclared. A separate prompt-disjoint sample of 512 responses from 128 prompts reports standardized reference-policy shift. It is diagnostic only and cannot accept or reject training.

B.5 COMPARATORS AND ABLATIONS

The ten trainable methods are:

1. **StableMax-PPO**: the complete method in Algorithm 1.
2. **Mean-PPO + Q**: PPO on μ_ω with the same mean, Quality, and KL protocol.
3. **GRPO + Q**: group-relative standardized μ_ω credit with matched safeguards (Shao et al., 2024; Guo et al., 2025).
4. **Scalar Max@32-PPO + Q**: exact leave-one-out scalar max credit $[\mu_i - \max_{j \neq i} \mu_j]_+$ with matched safeguards (Tang et al., 2025; Walder & Karkhanis, 2025; Bagirov et al., 2025).
5. **Entropic-PPO + Q**: PPO on $\text{CE}_\tau = -\tau \log \sum_r p_{\omega,r} e^{-r/\tau}$, using $\tau = 1$.
6. **Nominal Ordinal EV-PPO + Q**: exact J_{32}^{nom} without the calibration ambiguity set.
7. **StableMax-PPO without mean floor**: ablates nominal-mean non-inferiority.
8. **StableMax-PPO without Quality RM**: ablates the independent quality safeguard and is excluded from constrained-feasibility claims.
9. **Gaussian EV-PPO + Q**: the previous heteroscedastic Gaussian expected-maximum objective, retained as a model-assumption ablation.
10. **Top-4 PPO + Q**: exact optimization of $\Pr\{\exists j : R_j = 4\}$, isolating the highest threshold from the complete ordinal maximum.

Table 2: Complete aggregate table. Values are three-seed robust expected maximum, with 95% confidence intervals, held-out KL, and rollback counts.

Method	Robust max	95% CI	Held-out KL	Rollbacks/900
StableMax-PPO	3.458	[3.411, 3.506]	0.034	279 (31.0%)
Mean-PPO	3.424	[3.354, 3.517]	0.070	527 (58.6%)
GRPO	3.555	[3.530, 3.578]	0.094	741 (82.3%)
Scalar max-PPO	3.517	[3.476, 3.558]	0.054	285 (31.7%)
Entropic PPO	3.400	[3.353, 3.447]	0.037	302 (33.6%)
Nominal EV-PPO	3.421	[3.377, 3.467]	0.028	121 (13.4%)
StableMax-PPO (no mean floor)	3.508	[3.426, 3.574]	0.048	230 (25.6%)
StableMax-PPO (no quality floor)	3.467	[3.402, 3.530]	0.048	329 (36.6%)
Gaussian EV-PPO	3.558	[3.533, 3.581]	0.061	367 (40.8%)
Top-4 PPO	3.418	[3.353, 3.499]	0.042	277 (30.8%)
Best-of-32 (base)	3.086	[3.045, 3.125]	—	—

Table 3: Per-seed primary endpoints at $N = 32$. Each cell reports robust expected maximum / held-out KL on the same 512-prompt evaluation split. The KL values are the common-base-trajectory means from the held-out evaluator, matching the aggregate endpoint in Table 2.

Method	Seed 314	Seed 2718	Seed 1618
StableMax-PPO	3.503 / 0.057	3.452 / 0.024	3.420 / 0.022
Mean-PPO	3.369 / 0.070	3.523 / 0.074	3.381 / 0.066
GRPO	3.563 / 0.094	3.547 / 0.097	3.553 / 0.091
Scalar max-PPO	3.511 / 0.052	3.555 / 0.080	3.484 / 0.031

An eleventh, evaluation-only **Best-of-32** row samples 32 responses from the untrained base policy and ranks them by μ_ω . It is an inference-time reference, not a compute-matched training baseline; its generation and scoring cost is reported separately.

The comparator set is organized to separate three changes that are otherwise easy to conflate: the deployment functional (mean, scalar maximum, top-rating, or full ordinal maximum), the calibration correction, and the stabilizers. The nominal ordinal and Gaussian variants hold the policy optimizer fixed while changing the evaluator assumption; the no-floor variants hold the target fixed while removing one safeguard. This is why the ablations support a causal interpretation of the observed frontier without claiming that any single component is universally beneficial.

The per-seed view makes the operating-point claim testable without relying on the aggregate interval alone: StableMax-PPO has lower held-out KL than Mean-PPO and GRPO in every seed, while Scalar max-PPO is lower in two of the three seeds. Its robust maximum remains competitive across the same paired prompts. The robust values are the checked-in per-seed CSV records and the KL values are the held-out common-base means; all entries are rounded only for display, with no additional aggregation or seed selection performed here.

B.6 ABLATION AND STABILITY ANALYSIS

The component rows in Table 2 isolate the operating point controls. Removing the mean floor increases the aggregate robust maximum from 3.458 to 3.508, but also raises held-out KL from 0.034 to 0.048. Removing the independent Quality floor gives (3.467, 0.048), while removing calibration gives (3.421, 0.028), in (robust maximum, KL) coordinates. The three-seed intervals overlap and the per-seed effects are heterogeneous, so these are sensitivity comparisons rather than isolated causal effects. Calibration removal changes the evaluator-defined target; floor removal changes the accepted-update trajectory. The records therefore support a controlled frontier interpretation, not a claim that any safeguard is a universally improving reward term.

B.7 EVALUATION ENDPOINTS

Every method is evaluated on 512 frozen test prompts with newly generated groups of 32 responses. The primary paired prompt-level endpoint is h_{32}^{rob} , exactly matching the StableMax-PPO training objective. Secondary model-based endpoints are h_{32}^{nom} , U_{32}^{nom} , candidate-level μ_ω , ordinal variance, $p_{\omega,4}$, independent Quality score, reference KL, constraint residuals, accepted updates, and exact rollback counts.

The confirmatory evaluation fixes $N = 32$ for every method; no cross-budget sensitivity result is claimed. The group-level maximum uses all 32 candidates and does not require a selected response. A deployment-style $\arg \max_i \mu_\omega(x, Y_i)$ response is reported only as a secondary selector diagnostic. Held-out KL is computed on common base trajectories, so the paired comparison measures policy motion under the same prompt protocol rather than differences in test-set difficulty.

For completeness, the top-category diagnostic used in the cached evaluation is

$$U_N^{\text{nom}}(\theta) := \mathbb{E}_{x, Y_{1:N}} \left[1 - \prod_{j=1}^N (1 - p_{\omega, 4}(x, Y_j)) \right]. \quad (7)$$

It is the frozen-model probability that at least one candidate receives rating four. It is not a Gaussian tail approximation and is not the primary estimand; the complete ordinal maximum remains the reported objective.

B.8 STATISTICAL ANALYSIS AND SUCCESS CRITERIA

All seeds are reported separately. Aggregate uncertainty is estimated by a hierarchical bootstrap that resamples seeds and prompts. Paired prompt-level contrasts additionally report one-sided sign-flip randomization tests and matched-pairs rank-biserial effect sizes. The primary superiority gates require the lower endpoint of a two-sided 95% hierarchical-bootstrap interval for the StableMax-PPO improvement in h_{32}^{rob} to exceed zero against Mean-PPO, GRPO + Q, and Scalar Max@32-PPO.

Safeguard success requires the corresponding interval lower endpoints for StableMax-PPO’s candidate nominal mean and independent Quality mean to remain at or above m_0 and q_0 . A point estimate alone cannot satisfy either floor. The complete scientific-success decision requires all predeclared superiority, floor, and generated-output proxy gates; improvement under the target RM alone is insufficient.

B.9 ANTI-EXPLOITATION AND HUMAN-GROUNDED DIAGNOSTICS

Newly generated outputs are compared with Mean-PPO outputs by two frozen evaluators: the Qwen2.5-14B-Instruct judge and RLHF/low/ArmoRM-Llama3-8B-v0.1 (Zheng et al., 2023; Wang et al., 2024a). For each seed, the diagnostic aligns up to 512 held-out prompt groups, retains the response selected by $\arg \max_i \mu_\omega(x, Y_i)$ from each 32-response group, and evaluates the matched pair. Qwen judgments use randomized presentation order and a reverse order check; ArmoRM supplies an independent pointwise score. Each evaluator must first pass its frozen RewardBench calibration gate (between 512 and 1,024 calibration pairs, with the frozen split and hashes recorded by the protocol). Pairwise judgments are model-based proxies, not human labels. The complete group-level diagnostic is run separately on all 32 candidates and reports robust/nominal maximum, probability of any rating-4 response, and the paired group delta; selected-response diagnostics additionally report length, character count, repeated-trigram rate, unique-word ratio, refusal behavior, and upper-tail strata.

Table 4: Independent generated-output proxy comparisons against Mean-PPO. Preference is tie-adjusted, so 0.5 denotes parity; intervals are three-seed hierarchical-bootstrap 95% intervals.

Evaluator	Preference	95% CI
Qwen2.5-14B judge	0.494	[0.478, 0.510]
ArmoRM-Llama3-8B-v0.1	0.502	[0.458, 0.544]

Both scores are near parity and both intervals contain 0.5, so neither evaluator shows a clear quality-degradation signal relative to Mean-PPO. A target-RM increase accompanied by proxy degradation would be evidence consistent with reward-model exploitation; these results do not show that signature, but the model-based design cannot establish its absence.

RewardBench and Skywork chosen–rejected lockboxes contain previously collected human labels. For each policy, length-normalized chosen–rejected policy log-probability accuracy and margin are reported as preference-retention diagnostics. They are not Best-of-32 endpoints and do not label

newly sampled policy responses. In particular, a new response cannot be declared human preferred merely because it was generated for a prompt that also appears in a fixed preference benchmark.

No new human annotation is collected in this protocol. Therefore, even if all anti-exploitation gates pass, the permitted claim is *no evidence of exploitation under the measured model-based and existing-label diagnostics*. The experiment cannot establish absence of reward hacking or direct human preference for generated outputs. A blinded group-rating packet is emitted so that fresh human evaluation can be added later without regenerating or selecting outputs.

B.10 REPRODUCIBILITY AND REPORTING

Code, model adapters, data partitions, floor samples, calibration reports, and protocol files are content-hashed before training. A real-GPU smoke test must execute every declared objective, verify finite applicable metrics, verify constraint activation, and test exact parameter and optimizer rollback. Formal training and evaluation run only after these checks pass. Technical failures are versioned and corrected with regression tests. Unfavorable scientific outcomes are reported without changing seeds, thresholds, comparators, evaluators, or success criteria.

The 1.5B policy is evaluated by 14B reward and proxy models. This capacity gap may improve discrimination but may also create systematic evaluator shift. The three-seed design measures optimization variability; it does not replace replication at other model scales or fresh blinded labels.

C IMPLEMENTATION DETAILS FOR STABLEMAX-PPO

This appendix records implementation conventions for Algorithm 1, rather than a second algorithm. Appendix B specifies models, seeds, and thresholds; the machine-readable supplement provides aggregate and per-seed tables, terminal records, and SHA-256 hashes.

For each prompt x_b , the implementation caches behavior-policy token log probabilities and response masks before evaluating any reward model. The five ordinal probabilities are converted to prefix CDFs. For $t = 0, \dots, 3$, the robust envelope is $\bar{F}_{b,i}(t) = \min\{1, F_{\omega,b,i}(t) + \varepsilon_{\text{cal}}\}$, with $\bar{F}_{b,i}(-1) = 0$ and $\bar{F}_{b,i}(4) = 1$. The robust probabilities are adjacent differences of this envelope. Prefix and suffix products compute all group values and leave-one-out credits in $O(BN|\mathcal{R}|)$ time without enumerating rating tuples.

The constrained credit used by the PPO surrogate is $\tilde{q}_{b,i} = q_{b,i}^{\text{rob}} + \lambda_m \mu_\omega(x_b, Y_{b,i})/N + \lambda_q Q_\eta(x_b, Y_{b,i})/N$, with $A_{b,i} = N[\tilde{q}_{b,i} - b_\psi(x_b)]$. The prompt baseline is fitted only from accepted transitions. Dual variables use projected updates after acceptance; rejected steps leave all controller state unchanged. The lagged RMS scale is computed from accepted credits only, so a rejected outlier cannot change the next normalization factor.

Before the optimizer step, the implementation snapshots all trainable policy parameters and the complete optimizer state, including momentum and adaptive variance buffers. It evaluates full-categorical token KL on the same rollout states after the tentative step. If the hard threshold is exceeded, both snapshots are restored and no baseline, dual, KL-controller, or RMS state is advanced. This is an exact batch-level rollback invariant; it does not imply a population KL bound.

All floating-point reductions use deterministic tensor order where supported. The supplementary code contains tests for ordinal identities, credit reductions, finite metrics, protocol hashes, and rollback restoration.

Table 5: Implementation invariants checked by the artifact.

Component	Invariant	Test coverage
Ordinal kernel	Five probabilities form a valid distribution; CDF boundaries and threshold identities hold.	evrl_ordinal
Credit kernel	Prefix/suffix products match the exact credit and the $N = 1$ reduction.	evrl_reduction
Rollback state	Policy parameters and optimizer buffers are restored together after rejection.	publication.v8
Controller state	Baseline, dual, KL, and RMS states advance only on accepted updates.	publication.v8
Protocol integrity	Finite metrics, content hashes, and verification records are checked before aggregation.	v8 verification